Surface Gravity Waves

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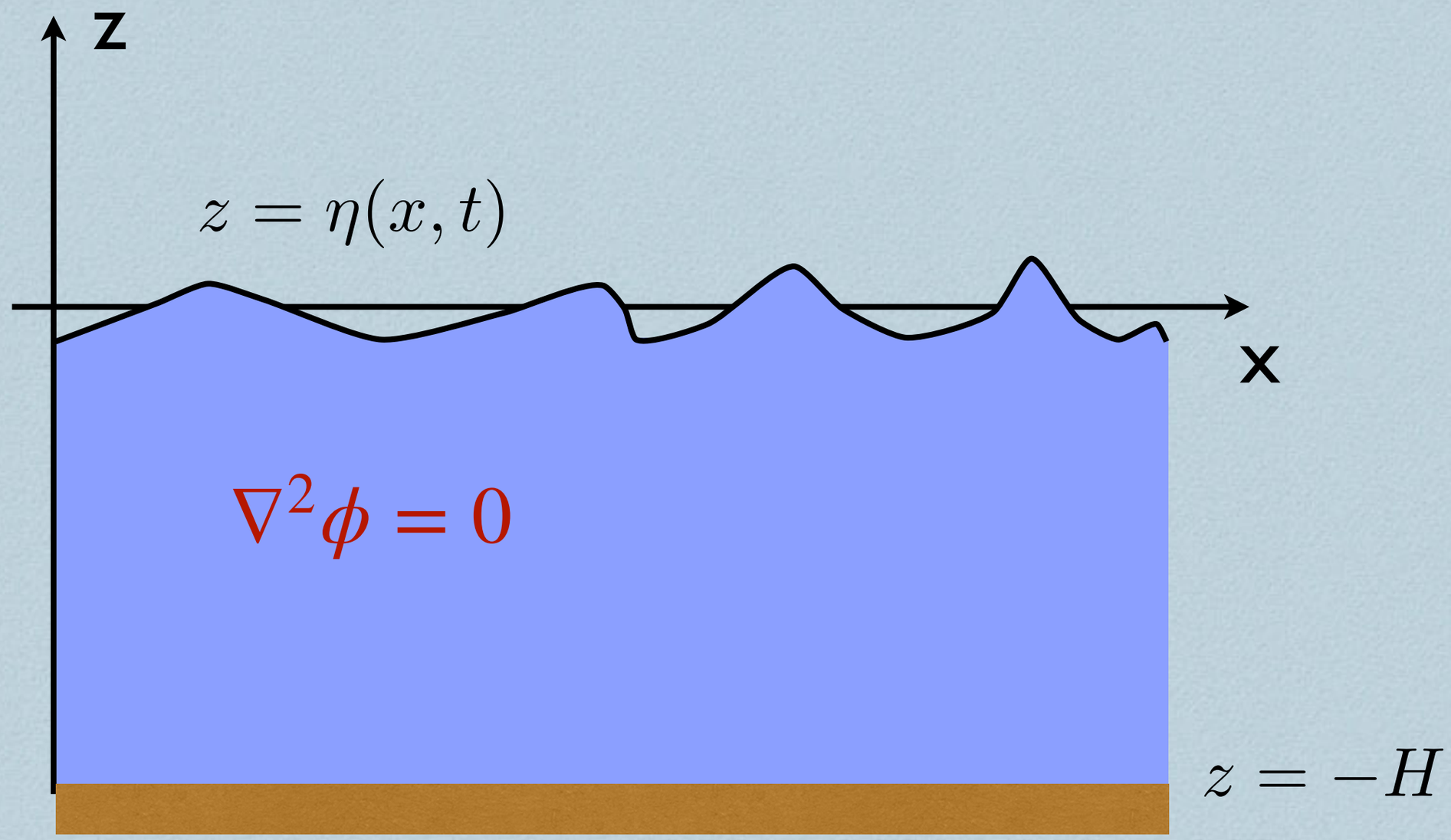
The Surface Wave Problem

- Surface waves are a readily observable phenomenon with interesting physics.
- Surface waves are horizontally isotropic (properties do not depend on the direction the wave number vector (k,l)).
- Waves of different wavelength are observed to propagate with different speeds (i.e. surface waves are dispersive).
- For a first attempt at a theory we want to neglect finite amplitude effects, and hence we linearize and use potential flow.
- However this still means we have to deal with the moving boundary.

The surface wave problem posed: just the equations

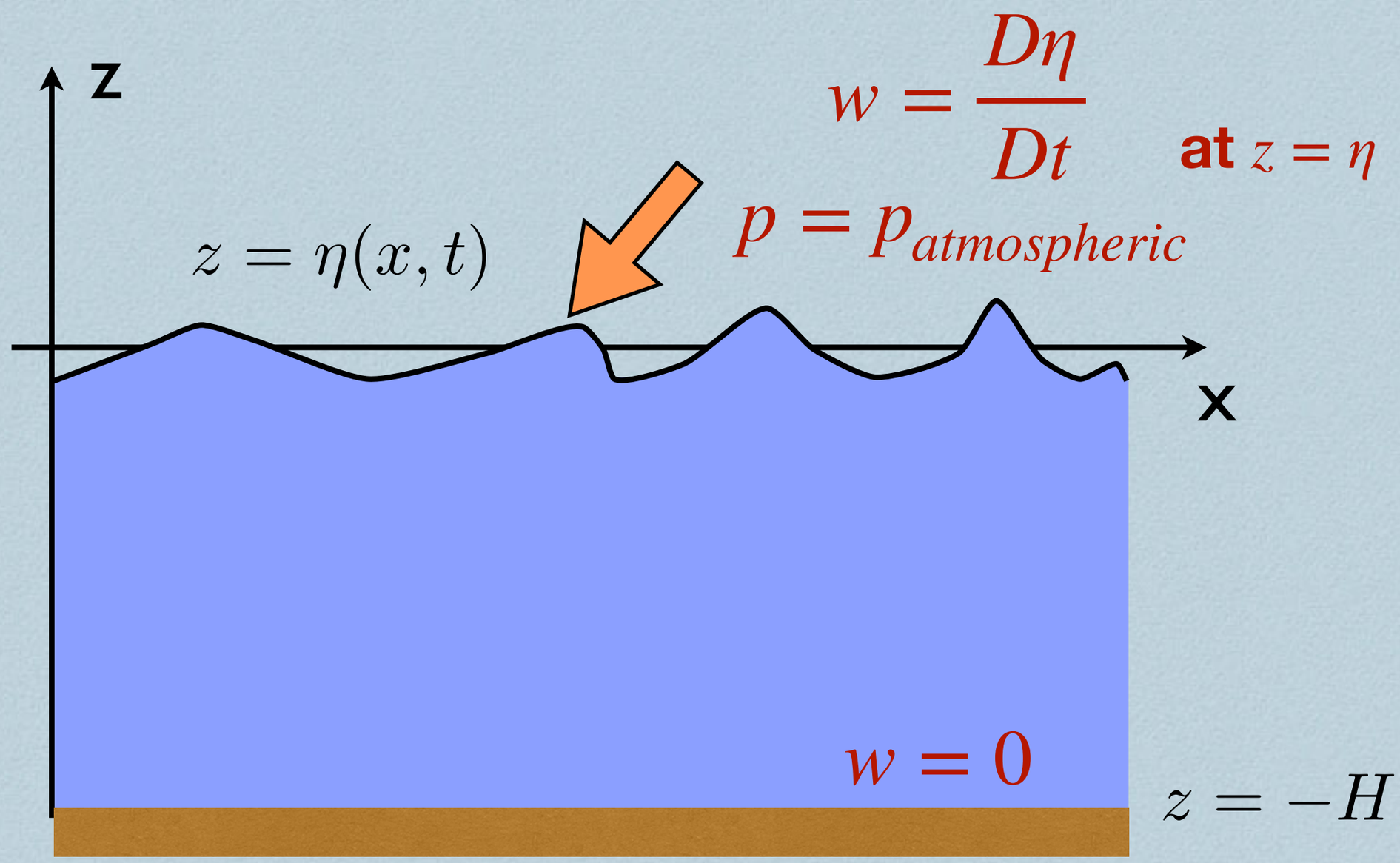
$$\vec{u} = \vec{\nabla}\phi$$



Deep Water Approximation takes

$$H \to \infty$$

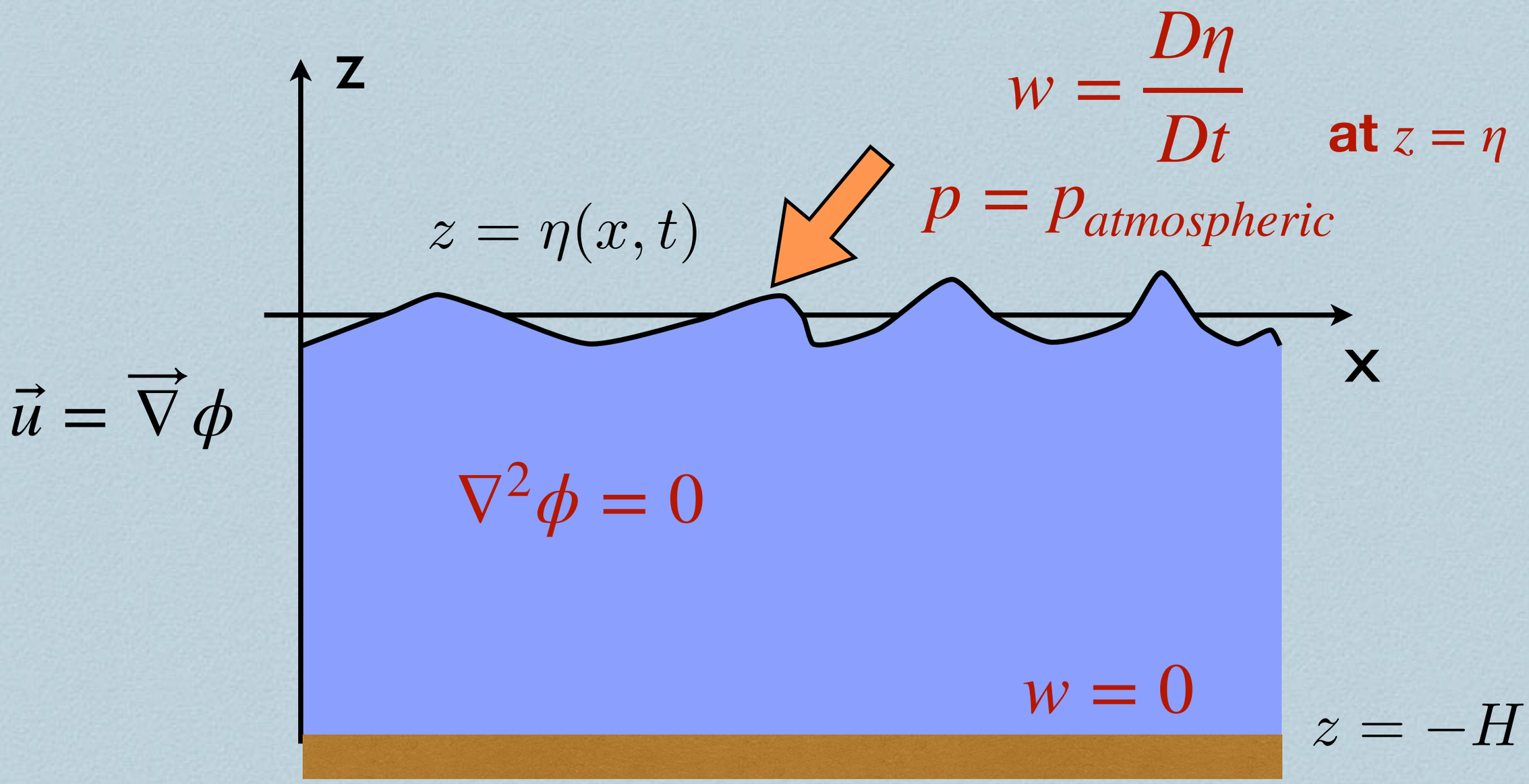
The surface wave problem posed: just the BCs



Deep Water Approximation takes

$$H \to \infty$$

The surface wave problem posed: all together



Deep Water Approximation takes

$$H \to \infty$$

Analytical progress is IMPOSSIBLE without approximation

Deriving the kinematic (or Las Vegas) condition

Start with the observation that fluid particles on the surface stay there (hence Vegas…) and write down the equation for the surface:

$$z = \eta(x, y, t)$$

And now take the material derivative of both sides:

$$\frac{Dz}{Dt} = \frac{D\eta}{Dt} \implies w = \frac{D\eta}{Dt}$$

Because:

$$\frac{Dz}{Dt} = \frac{\partial z}{\partial t} + u\frac{\partial z}{\partial x} + v\frac{\partial z}{\partial y} + w\frac{\partial z}{\partial z} = 0 + 0 + 0 + w = w$$

Linearizing the kinematic (or Las Vegas) condition

$$\frac{Dz}{Dt} = \frac{D\eta}{Dt} \implies w = \frac{D\eta}{Dt} \quad \textbf{at } z = \eta$$

We would like to approximate for small eta:

There are two parts to this:
1) drop the nonlinear terms in the material derivative
2) Apply the BC at z=0

$$w = \frac{\partial \eta}{\partial t} \qquad \textbf{at } z = 0$$

Bernoulli for surface waves

$$\frac{\partial \phi}{\partial t} + \frac{1}{2}(u^2 + w^2) + \frac{p}{\rho_0} + gz = F(t)$$

- Next absorb the F(t) term into the potential. **This is OK because only the gradient of the potential is physical.**

- Also for small waves can drop the nonlinear terms.

$$\frac{\partial \phi}{\partial t} + \frac{p}{\rho_0} + gz = 0$$

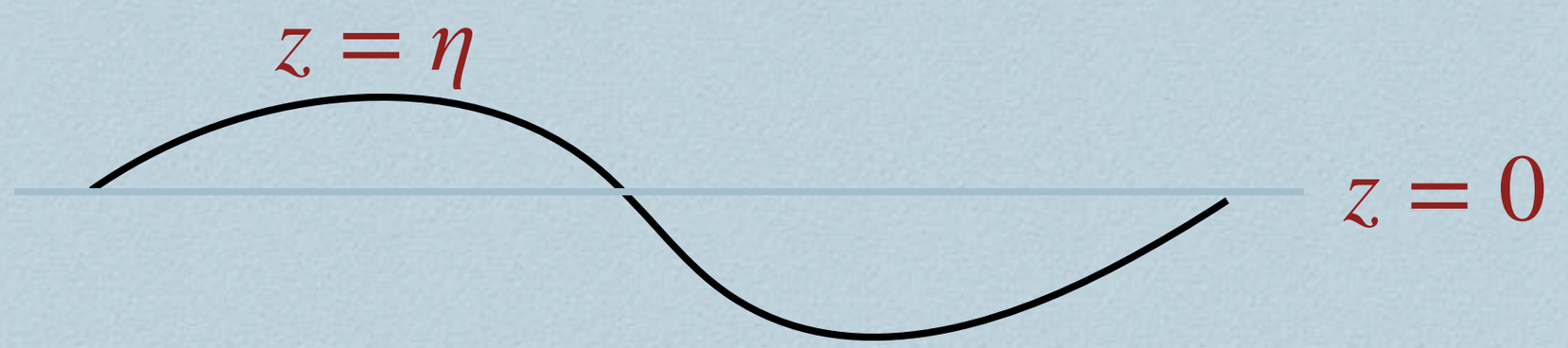
$$p = p_{atmosphere} \approx 0$$

$$\frac{\partial \phi}{\partial t} + gz = 0$$

- Technically these are applied at the unknown free surface, z=eta.

- But the small wave assumption should allow us to approximate to get a condition at z=0.

Deriving the linearized dynamic condition



$$f(\eta) = f(0) + \eta f'(0) + O(\eta^2)$$

$$\frac{\partial \phi}{\partial t} + gz = 0 \qquad at \qquad z = \eta$$

$$\boxed{\frac{\partial \phi}{\partial t} + g\eta = 0 \qquad at \qquad z = 0}$$

Notice that only phi and eta appear

The linearized surface wave problem statement and work

$$\nabla^2 \phi = 0$$

$$\frac{\partial \phi}{\partial t} + g\eta = 0 \quad \text{and} \quad \phi_z = \frac{\partial \eta}{\partial t} \quad \text{at} \quad z = 0$$

$$\phi_z \to 0 \quad \text{as} \quad z \to -\infty$$

Try $\eta(x,t) = \cos[kx - \sigma(k)t]$

$$\frac{\partial \phi}{\partial t} + g\eta = 0 \quad \text{implies} \quad \phi_t(x,0,t) = -g\cos[kx - \sigma(t)t]$$

So try $\phi(x,t) = f(z)\sin[kx - \sigma(k)t]$

$$\frac{d^2 f}{dz^2} - k^2 f = 0 \implies f(z) = c_1 \exp(kz) + c_2 \exp(-kz)$$

The linearized surface wave problem statement and work 2

$$\nabla^2 \phi = 0$$

$$\frac{\partial \phi}{\partial t} + g\eta = 0 \quad \text{and} \quad \phi_z = \frac{\partial \eta}{\partial t} \quad \text{at} \quad z = 0$$

$$\phi_z \to 0 \quad \text{as} \quad z \to -\infty$$

$$\frac{d^2 f}{dz^2} - k^2 f = 0 \implies f(z) = c_1 \exp(kz) + c_2 \exp(-kz)$$

$$\phi_z \to 0 \quad \text{as} \quad z \to -\infty \quad \text{gives} \quad f(z) = c_1 \exp(kz)$$

$$\phi_z = \frac{\partial \eta}{\partial t} \quad \text{at} \quad z = 0 \quad \text{gives} \quad c_1 = \frac{\sigma}{k}$$

$$\phi_t + g\eta = 0 \quad \text{at} \quad z = 0 \quad \text{gives} \quad \sigma^2 = gk$$

The linearized surface wave problem solution

$$c_p = \frac{\sigma}{k} \qquad \eta(x,t) = \cos[kx - \sigma(k)t] \qquad c_g = \frac{d\sigma}{dk}$$

$$\phi(x,t) = \frac{\sigma}{k}\sin[kx - \sigma(k)t]\exp(kz)$$

$$\sigma = \pm\sqrt{gk} \qquad c_p = \pm\sqrt{\frac{g}{k}} \qquad c_g = \pm\frac{1}{2}\sqrt{\frac{g}{k}}$$

- The waves are dispersive because c_p and c_g different. Longer waves travel faster.
- The velocities decay exponentially with depth.
- In the notes I allow eta to have an amplitude a_o, but here I used linearity to just set this constant to be equal to one.
- The deep water assumption made the algebra easier by letting us drop one of the two exponentials right away.

- The basic solution algorithm goes something like this:

1. Use an assumption about the nature of the velocities to simplify to a linear governing equation.

2. Write down the boundary conditions at the free surface and linearize them for the case of small waves.

3. Propose a plane wave solution and use this to separate variables in Laplace's equation.

4. Use the boundary conditions in the deep to simplify the solution for the potential.

5. Use the kinematic, or Las Vegas, boundary condition to solve for the constant of integration in the potential.

6. Use the dynamic boundary condition to solve for the dispersion relation.

The linearized surface wave problem solution

$$\eta(x,t) = \cos[kx - \sigma(k)t]$$

$$\phi(x,t) = \frac{\sigma}{k}\sin[kx - \sigma(k)t]\exp(kz)$$

$$\sigma = \pm\sqrt{gk} \qquad c_p = \pm\sqrt{\frac{g}{k}} \qquad c_g = \pm\frac{1}{2}\sqrt{\frac{g}{k}}$$

- The form of the solution is explicit so we could try to find pathlines.
- If we assume we can linearize the path lines are circular (see notes).
- But if you've thrown a stick in a wavy lake you know there is a small drift. This is called Stokes Drift and is due to the fact that we can't quite linearize.
- https://www.youtube.com/watch?v=oPplzgEcyGc
- https://www.youtube.com/watch?v=TTNbjXE19Ss
- https://www.youtube.com/watch?v=WffR6HrEqTA